Project Development Phase — Story Design (Narrative Presentation)

Date	_____
Team ID	_____
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis

1. What is the Story View?

The Story view is a 10-slide guided narrative published on Tableau Public. Unlike the free-exploration dashboard where users choose their own path, the Story view takes a single viewer through a structured, opinionated presentation of the most important EV insights — in a logical sequence from problem to data to finding to action.

The Story view serves three purposes: (1) it makes the project accessible to an audience who does not want to explore data themselves; (2) it provides a polished, shareable presentation link for classroom or professional settings; (3) it demonstrates that the team understands not just how to visualise data, but how to communicate data insights effectively.

2. Story Narrative Structure

The story follows a classic problem-finding-action narrative arc:

- Slides 1–3: Establish the Problem and the Data (Why does this matter? What data do we have?)
- Slides 4–7: The Key Findings (What does the data actually show? Four major insights.)
- Slides 8–9: Takeaways and Actions (What should drivers and fleet managers do differently because of this?)
- Slide 10: Future Vision (What comes next for this tool and for the EV data ecosystem?)

3. Slide-by-Slide Design Brief

Slide	Title	Content and Design Brief
1	**The EV Range Problem**	Opening slide — bold statement: "Your car says 400 km. Real life says 280 km. Here is why." Project title and team displayed. Sets up the central tension between advertised and real-world EV range.
2	**Who This Affects**	Three user persona cards: EV Driver (range anxiety), Fleet Manager (charging costs), Researcher (data access). Brief empathy statements for each. Shows the breadth of people affected by the problem.
3	**Our Dataset**	Dataset summary: 10,000+ real-world EV sessions, 15 models, 4 seasons, 3 terrain types, data from multiple countries. Key stat callouts: number of records, date range, models covered. Establishes data credibility.
4	**Real Range vs. Advertised Range**	Bar chart comparing advertised range (from manufacturer spec) with average real-world range for each of the 15 models. Key finding: real range averages 18–22% lower than advertised across all models.

5	**How Temperature Kills Your Range**	Scatter chart: temperature vs. range. Annotated with two key thresholds: "Range drops 15% below +5°C" and "Range drops 30% below -10°C." Season colour-coding makes the pattern instantly clear.
6	**How Fast Do EVs Actually Charge?**	Charge curve animation showing battery % vs. time for three representative models. Annotated: "0–80% takes 28 minutes. 80–100% takes 45 minutes." Reveals the fast-charge plateau that most EV buyers do not know about.
7	**Which Routes Drain the Battery Most?**	Geographic heatmap showing energy consumption intensity by route type and terrain. Key finding: hilly routes consume 22% more energy than flat routes at the same speed. City routes have more stop-start loss than highway.
8	**Battery Ageing — What to Expect**	Degradation trend line chart: range over cumulative charge cycles for four models. Key finding: best-performing models retain 92% range at 500 cycles; worst retain 78%. Fleet managers can use this for battery replacement planning.
9	**Key Takeaways for Drivers and Fleet Managers**	Summary slide with three action-oriented insights: (1) Add 25% buffer to advertised range in winter planning. (2) Charge to 80% for daily use; full charge only before long trips. (3) Model X retains range best for high-mileage fleet use.
10	**What Comes Next**	Future scope slide: Live OBD-II data integration, AI-based range prediction, mobile app, charging station live availability overlay, global EV model expansion. Team credits and Tableau Public dashboard link.

4. Story Design Principles

- Each slide must have one central insight — not multiple competing messages.
- Every chart is annotated with a plain-English finding directly on the visual — not just in a caption below.
- Slides use large typography for the key number or finding; supporting text is minimal.
- Colour is used consistently: red = bad/critical, amber = caution, green = good. This is maintained across all 10 slides.
- The story should be understandable without any audio or presenter — it must stand alone as a written visual narrative.
- Each slide ends with a one-sentence "So what?" that tells the viewer what to do with this insight.

5. Publishing the Story

1. Build all charts in Tableau Desktop or Tableau Public Desktop.
2. Assemble the 10 story slides in the Tableau Story feature.
3. Connect each slide to the relevant chart or sheet.
4. Add annotations and plain-English text boxes directly on each slide.
5. Click "Publish to Tableau Public" and set visibility to Public.
6. Copy the Tableau Public link and add it to the Product Documentation (P7_02) and Final Report (P7_01).